

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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### COURSE OUTCOME COVERED

**CO1:** Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2  
Assessment Tool Weightage: 30%

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**QUESTIONS: 2**

**Total Marks:50**

Annexure A

Technical Presentation

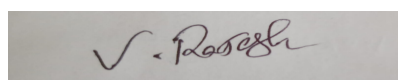
| <b>Criteria</b>  | <b>Technical Content Accuracy(3)</b> | <b>Problem Identification(2.5)</b> | <b>Use of Tools(2)</b> | <b>Communication(1.5)</b> | <b>Professionalism(1)</b> | <b>Total(10)</b> |
|------------------|--------------------------------------|------------------------------------|------------------------|---------------------------|---------------------------|------------------|
| <b>Weightage</b> | 30%                                  | 25%                                | 20%                    | 20%                       | 10%                       | 100%             |
| <b>Q1</b>        |                                      |                                    |                        |                           |                           |                  |
| <b>Q2</b>        |                                      |                                    |                        |                           |                           |                  |

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### **Code of Conduct:**

*I (192311037 / V.Rajesh ) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*



**Signature of the student:**

## RUBRICS FOR CALCULATION:

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10        | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness    |

## Annexure A

### Topic 1: Decision Support Systems (DSS)

#### Focus Areas:

- Definition and need of DSS
  - Components (Data, Model, UI)
  - Types of DSS
  - Real-world applications (banking, healthcare)
  - Advantages & limitations
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### Topic 2: Operational Systems vs DSS

#### Focus Areas:

1. OLTP vs OLAP concepts
  2. Key differences (data, users, purpose)
  3. Performance comparison
  4. Use-case scenarios
  5. Integration between systems
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### Topic 3: Architecture of Data Warehouse

#### Focus Areas:

1. Data warehouse definition
  2. Three-tier architecture
  3. ETL process
  4. Data marts & metadata
  5. Benefits in decision-making
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## Topic 4: OLAP and Data Cubes

### Focus Areas:

1. OLAP concept
  2. Data cube structure
  3. OLAP operations (slice, dice, roll-up, drill-down)
  4. Applications in analytics
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## Topic 5: Dimensional Data Modeling

### Focus Areas:

1. Fact and dimension tables
  2. Star schema vs snowflake schema
  3. Schema design process
  4. Real-world example
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## Topic 6: Query Processing & BI Tools

### Focus Areas:

1. Query processing stages
  2. Query optimization
  3. BI tools overview
  4. Reporting and visualization
  5. Examples of open-source BI tools
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## Topic 7: OLAP Application using KNIME

### Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation

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## PART B: PRESENTATION RULES & GUIDELINES

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### ◇ 1. Group Formation

1. Each group: 3–5 students
2. Each member must participate
3. Assign roles:
  1. Presenter
  2. Designer (PPT)
  3. Researcher
  4. Demo/Case handler

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### ◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
  1. Definition and importance
2. Core Concepts
  1. Explanation with diagrams
3. Process/Architecture
  1. Step-by-step explanation
4. Real-World Example
  1. Case study or application
5. Tool/Demo (if applicable)
  1. KNIME or BI tool
6. Conclusion
  1. Summary + future scope

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### ◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
  1. Diagrams
  2. Flowcharts

3. Tables (for comparison)
  3. Avoid:
    1. Too much text
    2. Copy-paste content
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## ◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
  1. Eye contact
  2. Clear voice
  3. Logical flow



# WHAT IS ETL?

**“EXTRACT ... TRANSFORM ... LOAD”**

CSA1603 - DATA WAREHOUSING AND DATA MINING

192311037

VAYILA RAJESH

# WHY ETL?

- ❖ Companies need a way to analyze their data for critical business decisions.
- ❖ Transactional Database can't answer complex business questions.
- ❖ A Data Warehouse provides a common data repository.
- ❖ ETL integrates data from multiple sources into a single unified system.
- ❖ ETL improves data quality by removing duplicate, inconsistent, and incorrect data.
- ❖ ETL prepares clean and organized data for faster reporting, OLAP analysis, and Business Intelligence (BI).



# ETL CONCEPT

- ❖ A Company data may be scattered in different locations and in different formats.
- ❖ ETL allows you to migrate the data into a Data Warehouse.
- ❖ ETL converts various formats and data types into a common standard.
- ❖ ETL integrates data from multiple sources into a single, unified system.
- ❖ ETL cleans and validates data by removing duplicate, inconsistent, and incorrect records.
- ❖ ETL prepares high-quality data for reporting, OLAP, and Business Intelligence (BI) applications.

# EXTRACT

## ❖ **Gathering the data**

- ☐ Raw data that is written directly into the disk.
- ☐ Data written to flat files or relational tables from structured source systems.
- ☐ Data can be read multiple times, if needed.

## ❖ **Cleansing the data**

- ☐ Remove duplicate records.
- ☐ Correct missing or invalid values.
- ☐ Eliminate inconsistent and unwanted data before transformation

# TRANSFORM

- ❖ Preparing the data to be housed in the data warehouse.
- ❖ Converting the extracted data
  - ☐ Using rules and lookup tables
  - ☐ Combining data
  - ☐ Verification/Validity checks
  - ☐ Standardization

# LOAD

## ❖ **Storing the transformed data in the Data Warehouse**

- ☐ Loads clean and validated data into the target database.
- ☐ Ensures data is stored securely and consistently.
- ☐ Makes data ready for analysis and reporting.

## ❖ **Loading Methods**

- ☐ Supports Batch and Real-time processing.
- ☐ Can follow Star Schema and Snowflake Schema.

# ETL FLOW



## Extract

Retrieves and verifies data  
from various sources



## Transform

Processes and organizes  
extracted data so it is usable



## Load

Moves transformed data  
to a data repository



The background is a blue gradient. In the corners, there are decorative white lines resembling circuit traces or a stylized city skyline, with small circles at the end of the lines.

THANK YOU